

I. IDEMPOTENCY OF λ_1

$$\lambda_1 = \sum_{1 \leq l \leq r} \alpha_l a_l^\dagger a_l \quad (1)$$

$$(\lambda_1)^2 = \sum_{1 \leq l \leq r} \alpha_l^2 a_l^\dagger a_l + \sum_{1 \leq l_1 \neq l_2 \leq r} \alpha_{l_1} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} \quad (2)$$

$$(\lambda_1)^2 - \lambda_1 = \sum_{1 \leq l \leq r} \alpha_l (\alpha_l - 1) a_l^\dagger a_l + \sum_{1 \leq l_1 \neq l_2 \leq r} \alpha_{l_1} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} = 0$$

$$\Updownarrow$$

$$\lambda_1 = a_{l_0}^\dagger a_{l_0} \quad (3)$$

It is also useful to define

$$\Lambda_{111} = \sum_{1 \leq l \leq r} \alpha_l (\alpha_l - 1) a_l^\dagger a_l \quad (4)$$

and

$$\Lambda_{112} = 2 \sum_{1 \leq l_1 < l_2 \leq r} \alpha_{l_1} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} \quad (5)$$

so that idempotency is equivalent to

$$\Lambda_{111} + \Lambda_{112} = 0. \quad (6)$$

Linear independency gives

$$\alpha_l (\alpha_l - 1) a_l^\dagger a_l = 0 \forall l \quad (7)$$

and

$$\alpha_{l_1} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} = 0; 1 \leq l_1 < l_2 \leq r. \quad (8)$$

A. Relation to First Quantization

Noting that

$$\begin{aligned} \lambda_1 &= \Gamma \left(\hat{\lambda}_1 \right); \hat{\lambda}_1 \in \mathcal{B} \left(\mathcal{H}^1 \right) \\ \hat{\lambda}_1 &= \Gamma^{-1} \left(\lambda_1 \right) \end{aligned} \quad (9)$$

one obtains

$$\begin{aligned}
(\lambda_1)^2 &= \sum_{1 \leq l \leq r} \alpha_l^2 a_l^\dagger a_l + \sum_{1 \leq l_1 \neq l_2 \leq r} \alpha_{l_1} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} \\
&= \Gamma(\hat{\lambda}_1^2) + \Gamma(\hat{\lambda}_1 \wedge \hat{\lambda}_1) = \Gamma(\hat{\lambda}_1^2 + \hat{\lambda}_1 \wedge \hat{\lambda}_1),
\end{aligned} \tag{10}$$

thus

$$\begin{aligned}
(\lambda_1)^2 - \lambda_1 &= \Gamma(\hat{\lambda}_1^2 - \hat{\lambda}_1 + \hat{\lambda}_1 \wedge \hat{\lambda}_1) = 0 \\
&\Downarrow \\
\hat{\lambda}_1^2 - \hat{\lambda}_1 &= 0 \ \& \ \hat{\lambda}_1 \wedge \hat{\lambda}_1 = 0.
\end{aligned} \tag{11}$$

The condition

$$\hat{\lambda}_1^2 - \hat{\lambda}_1 = 0 \tag{12}$$

shows that $\hat{\lambda}_1$ has eigenvalues $\{0, 1\}$ and the condition

$$\begin{aligned}
\hat{\lambda}_1 \wedge \hat{\lambda}_1 = 0 &= \sum_{1 \leq l_1 < l_2 \leq r} \alpha_{l_1} \alpha_{l_2} |\varphi_{l_1}\rangle \langle \varphi_{l_1}| \wedge |\varphi_{l_2}\rangle \langle \varphi_{l_2}| = 2 \sum_{1 \leq l_1 < l_2 \leq r} \alpha_{l_1} \alpha_{l_2} |\varphi_{l_1} \wedge \varphi_{l_2}\rangle \langle \varphi_{l_1} \wedge \varphi_{l_2}| \\
&= \sum_{1 \leq l_1 < l_2 \leq r} \alpha_{l_1} \alpha_{l_2} |\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{l_1} \varphi_{l_2}|,
\end{aligned} \tag{13}$$

[as

$$|\varphi_{l_1} \varphi_{l_2}\rangle = \frac{1}{\sqrt{2}} |\varphi_{l_1} \wedge \varphi_{l_2}\rangle] \tag{14}$$

shows that

$$\text{rank} \left\{ \hat{\lambda}_1 \right\} = 1. \tag{15}$$

II. IDEMPOTENCY OF λ_2

The commutation relationship

$$\begin{aligned}
\left[a_{k_1}^\dagger a_{k_2}^\dagger, a_{l_2} a_{l_1} \right] &= I_{\mathcal{H}_{\mathcal{F}}} \delta_{l_2 k_1} \delta_{l_1 k_2} - I_{\mathcal{H}_{\mathcal{F}}} \delta_{l_2 k_2} \delta_{l_1 k_1} + a_{k_2}^\dagger a_{l_2} \delta_{l_1 k_1} \\
&\quad - a_{k_1}^\dagger a_{l_2} \delta_{l_1 k_2} - a_{k_2}^\dagger a_{l_1} \delta_{l_2 k_1} + a_{k_1}^\dagger a_{l_1} \delta_{l_2 k_2}
\end{aligned} \tag{16}$$

gives

$$\begin{aligned}
&a_{l_4} a_{l_3} a_{k_1}^\dagger a_{k_2}^\dagger \\
&= a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} - I_{\mathcal{H}_{\mathcal{F}}} \delta_{l_4 k_1} \delta_{l_3 k_2} + I_{\mathcal{H}_{\mathcal{F}}} \delta_{l_4 k_2} \delta_{l_3 k_1} - a_{k_2}^\dagger a_{l_4} \delta_{l_3 k_1} \\
&\quad + a_{k_1}^\dagger a_{l_4} \delta_{l_3 k_2} + a_{k_2}^\dagger a_{l_3} \delta_{l_4 k_1} - a_{k_1}^\dagger a_{l_3} \delta_{l_4 k_2}.
\end{aligned} \tag{17}$$

This can be used in

$$\begin{aligned}
\lambda_2^2 &= \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r \\ 1 \leq k_1 < k_2 \leq r \\ 1 \leq k_3 < k_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger \left(a_{l_4} a_{l_3} a_{k_1}^\dagger a_{k_2}^\dagger \right) a_{k_4} a_{k_3} \\
&= \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger \left\{ a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} - I_{\mathcal{H}\mathcal{F}} \delta_{l_4 k_1} \delta_{l_3 k_2} + I_{\mathcal{H}\mathcal{F}} \delta_{l_4 k_2} \delta_{l_3 k_1} \right. \\
&\quad \left. - a_{k_2}^\dagger a_{l_4} \delta_{l_3 k_1} + a_{k_1}^\dagger a_{l_4} \delta_{l_3 k_2} + a_{k_2}^\dagger a_{l_1} \delta_{l_4 k_1} - a_{k_1}^\dagger a_{l_3} \delta_{l_4 k_2} \right\} a_{k_4} a_{k_3} \\
&\quad \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} \left\{ a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} - a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \delta_{l_4 k_1} \delta_{l_3 k_2} \right. \\
&\quad + a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \delta_{l_4 k_2} \delta_{l_3 k_1} - a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_4} a_{k_4} a_{k_3} \delta_{l_3 k_1} + a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{l_4} a_{k_4} a_{k_3} \delta_{l_3 k_2} \\
&\quad \left. + a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_1} a_{k_4} a_{k_3} \delta_{l_4 k_1} - a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{l_1} a_{k_4} a_{k_3} \delta_{l_4 k_2} \right\}, \tag{18}
\end{aligned}$$

giving

$$\begin{aligned}
\lambda_2^2 &= \frac{1}{16} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq k_1, k_2 \leq r \\ 1 \leq k_3, k_4 \leq r}} \left\{ \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} \right. \\
&\quad - \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \delta_{l_4 k_1} \delta_{l_3 k_2} \\
&\quad + \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \delta_{l_4 k_2} \delta_{l_3 k_1} - \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_4} a_{k_4} a_{k_3} \delta_{l_3 k_1} \\
&\quad + \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{l_4} a_{k_4} a_{k_3} \delta_{l_3 k_2} \\
&\quad \left. + \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} \delta_{l_4 k_1} - \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{l_3} a_{k_4} a_{k_3} \delta_{l_4 k_2} \right\}, \tag{19}
\end{aligned}$$

which after contracting the deltas gives

$$\begin{aligned}
\lambda_2^2 &= \frac{1}{16} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq k_1, k_2 \leq r \\ 1 \leq k_3, k_4 \leq r}} \left\{ \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} - \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2l_4 l_3 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \right. \\
&\quad + \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2l_3 l_4 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} - \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2l_3 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_4} a_{k_4} a_{k_3} \\
&\quad + \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 l_3 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{l_4} a_{k_4} a_{k_3} \\
&\quad \left. + \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2l_4 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} - \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 l_4 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{l_3} a_{k_4} a_{k_3} \right\}, \tag{20}
\end{aligned}$$

permuting the field operators gives

$$\begin{aligned} \lambda_2^2 = \frac{1}{16} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq k_1, k_2 \leq r \\ 1 \leq k_3, k_4 \leq r}} \left\{ \lambda_{2l_1 l_2 l_3 l_4} \lambda_{2k_1 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} \right. \\ \left. + 4\lambda_{2l_1 l_2 l_3 l_4} \lambda_{2l_4 k_2 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} + 2\lambda_{2l_1 l_2 l_3 l_4} \lambda_{2l_3 l_4 k_3 k_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \right\} \end{aligned} \quad (21)$$

and ordering the indices leads to

$$\begin{aligned} \lambda_2^2 = 36 \sum_{\substack{1 \leq l_1 < l_2 < k_1 < k_2 \leq r \\ 1 \leq l_3 < l_4 < k_3 < k_4 \leq r \\ \sigma, \eta \in S_4}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)\eta(l_4)} \lambda_{2\sigma(k_1)\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} \\ + 9 \sum_{\substack{1 \leq l_1 < l_2 < k_2 \leq r \\ 1 \leq l_3 < k_3 < k_4 \leq r \\ 1 \leq l_4 \leq r}} \sum_{\sigma, \eta \in S_3} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)l_4} \lambda_{2l_4\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} \\ + \frac{1}{4} \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq k_3 < k_4 \leq r \\ 1 \leq l_3, l_4 \leq r}} \sum_{\sigma, \eta \in S_2} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)l_3 l_4} \lambda_{2l_3 l_4 \eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3}. \end{aligned} \quad (22)$$

Hence

$$\lambda_2^2 - \lambda_2 = \Lambda_{224} + \Lambda_{223} + \Lambda_{222}, \quad (23)$$

where

$$\Lambda_{224} = 36 \sum_{\substack{1 \leq l_1 < l_2 < k_1 < k_2 \leq r \\ 1 \leq l_3 < l_4 < k_3 < k_4 \leq r \\ \sigma, \eta \in S_4}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)\eta(l_4)} \lambda_{2\sigma(k_1)\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3}, \quad (24)$$

$$\Lambda_{223} = 9 \sum_{\substack{1 \leq l_1 < l_2 < k_2 \leq r \\ 1 \leq l_3 < k_3 < k_4 \leq r \\ \sigma, \eta \in S_3 \\ 1 \leq l_4 \leq r}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)l_4} \lambda_{2l_4\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} \quad (25)$$

and

$$\begin{aligned} \Lambda_{222} &= \frac{1}{4} \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq k_3 < k_4 \leq r}} \sum_{\substack{\sigma, \eta \in S_2 \\ 1 \leq l_3, l_4 \leq r}} \left\{ (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)l_3 l_4} \lambda_{2l_3 l_4 \eta(k_3)\eta(k_4)} - \lambda_{2l_1 l_2 k_3 k_4} \right\} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \\ &= \frac{1}{4} \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq k_3 < k_4 \leq r}} \sum_{\sigma, \eta \in S_2} \left\{ (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(k_3)\eta(k_4)}^2 - \lambda_{2l_1 l_2 k_3 k_4} \right\} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \\ &= \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq k_3 < k_4 \leq r}} \left\{ \lambda_{2l_1 l_2 k_3 k_4}^2 - \lambda_{2l_1 l_2 k_3 k_4} \right\} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3}. \end{aligned} \quad (26)$$

Idempotency is equivalent to

$$\Lambda_{224} + \Lambda_{223} + \Lambda_{222} = 0. \quad (27)$$

Linear independency gives for

- four particle operators

$$\sum_{\sigma, \eta \in S_4} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(i_1)\sigma(i_2)\eta(k_1)\eta(k_2)} \lambda_{2\sigma(i_3)\sigma(i_4)\eta(k_3)\eta(k_4)} a_{i_1}^\dagger a_{i_2}^\dagger a_{i_3}^\dagger a_{i_4}^\dagger a_{k_4} a_{k_3} a_{k_2} a_{k_1} = 0, \quad (28)$$

which gives

$$\sum_{\sigma, \eta \in S_4} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(i_1)\sigma(i_2)\eta(k_1)\eta(k_2)} \lambda_{2\sigma(i_3)\sigma(i_4)\eta(k_3)\eta(k_4)} = 0 \text{ or } a_{i_1}^\dagger a_{i_2}^\dagger a_{i_3}^\dagger a_{i_4}^\dagger a_{k_4} a_{k_3} a_{k_2} a_{k_1} = 0. \quad (29)$$

In particular

$$\begin{aligned} & \lambda_{1212}\lambda_{3434} + \lambda_{1213}\lambda_{3424} + \lambda_{1214}\lambda_{3423} + \lambda_{1223}\lambda_{3414} + \lambda_{1224}\lambda_{3413} + \lambda_{1234}\lambda_{3412} + \\ & \lambda_{1312}\lambda_{2434} + \lambda_{1313}\lambda_{2424} + \lambda_{1314}\lambda_{2423} + \lambda_{1323}\lambda_{2414} + \lambda_{1324}\lambda_{2413} + \lambda_{1334}\lambda_{2412} + \\ & \lambda_{1412}\lambda_{2334} + \lambda_{1413}\lambda_{2324} + \lambda_{1414}\lambda_{2323} + \lambda_{1423}\lambda_{2314} + \lambda_{1424}\lambda_{2313} + \lambda_{1434}\lambda_{2312} + \\ & \lambda_{2312}\lambda_{1434} + \lambda_{2313}\lambda_{1424} + \lambda_{2314}\lambda_{1423} + \lambda_{2323}\lambda_{1414} + \lambda_{2324}\lambda_{1413} + \lambda_{2334}\lambda_{1412} + \\ & \lambda_{2412}\lambda_{1334} + \lambda_{2413}\lambda_{1324} + \lambda_{2414}\lambda_{1323} + \lambda_{2423}\lambda_{1314} + \lambda_{2424}\lambda_{1313} + \lambda_{2434}\lambda_{1312} + \\ & \lambda_{3412}\lambda_{1234} + \lambda_{3413}\lambda_{1224} + \lambda_{3414}\lambda_{1223} + \lambda_{3423}\lambda_{1214} + \lambda_{3424}\lambda_{1213} + \lambda_{3434}\lambda_{1212} \end{aligned} \quad (30)$$

$$\begin{aligned} & 2\lambda_{1212}\lambda_{3434} + 2\lambda_{1213}\lambda_{3424} + 2\lambda_{1214}\lambda_{3423} + 2\lambda_{1223}\lambda_{3414} + 2\lambda_{1224}\lambda_{3413} + 2\lambda_{1234}\lambda_{3412} + \\ & 2\lambda_{1312}\lambda_{2434} + 2\lambda_{1313}\lambda_{2424} + 2\lambda_{1314}\lambda_{2423} + 2\lambda_{1323}\lambda_{2414} + 2\lambda_{1324}\lambda_{2413} + 2\lambda_{1334}\lambda_{2412} + \\ & 2\lambda_{1412}\lambda_{2334} + 2\lambda_{1413}\lambda_{2324} + 2\lambda_{1414}\lambda_{2323} + 2\lambda_{1423}\lambda_{2314} + 2\lambda_{1424}\lambda_{2313} + 2\lambda_{1434}\lambda_{2312} \\ & c_{i_1 i_2 i_3 i_4 k_1 k_2 k_3 k_4} a_{i_1}^\dagger a_{i_2}^\dagger a_{i_3}^\dagger a_{i_4}^\dagger a_{k_4} a_{k_3} a_{k_2} a_{k_1} = \{ \lambda_{2i_1 i_2 k_1 k_2} \lambda_{2i_3 i_4 k_3 k_4} + \} a_{i_1}^\dagger a_{i_2}^\dagger a_{i_3}^\dagger a_{i_4}^\dagger a_{k_4} a_{k_3} a_{k_2} a_{k_1} \end{aligned} \quad (31)$$

Conjecture

$$\Lambda_{224} = \Gamma(\lambda_2 \wedge \lambda_2) = 0 \quad (32)$$

- three particle operators

$$\begin{aligned}
& \sum_{\substack{\sigma, \eta \in S_3 \\ 1 \leq l_4 \leq r}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(k_1)l_4} \lambda_{2l_4\sigma(l_3)\eta(k_3)\eta(k_2)} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_3}^\dagger a_{k_1} a_{k_2} a_{k_3} = 0, \\
& \Updownarrow \\
& \sum_{\substack{\sigma, \eta \in S_3 \\ 1 \leq l_4 \leq r}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(k_1)l_4} \lambda_{2l_4\sigma(l_3)\eta(k_3)\eta(k_2)} = 0 \text{ or } a_{l_1}^\dagger a_{l_2}^\dagger a_{l_3}^\dagger a_{k_1} a_{k_2} a_{k_3} = 0
\end{aligned} \tag{33}$$

which gives

$$\begin{aligned}
& \sum_{\substack{\sigma, \eta \in S_3 \\ 1 \leq l_4 \leq r}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(k_1)l_4} \lambda_{2l_4\sigma(l_3)\eta(k_3)\eta(k_2)} = \\
& -4 (\lambda_{2l_1l_2k_1l_4} \lambda_{2l_4l_3k_2k_3} + \lambda_{2l_1l_2k_2l_4} \lambda_{2l_4l_3k_1k_3} + \lambda_{2l_1l_2k_3l_4} \lambda_{2l_4l_3k_1k_2}) \\
& -4 (\lambda_{2l_1l_3k_1l_4} \lambda_{2l_4l_2k_2k_3} + \lambda_{2l_1l_3k_2l_4} \lambda_{2l_4l_2k_1k_3} + \lambda_{2l_1l_3k_3l_4} \lambda_{2l_4l_2k_1k_2}) \\
& -4 (\lambda_{2l_2l_3k_1l_4} \lambda_{2l_4l_1k_2k_3} + \lambda_{2l_2l_3k_2l_4} \lambda_{2l_4l_1k_1k_3} + \lambda_{2l_2l_3k_3l_4} \lambda_{2l_4l_1k_1k_2}) = 0
\end{aligned} \tag{34}$$

In particular for diagonal terms

$$\begin{aligned}
& \lambda_{2l_1l_2l_1l_4} \lambda_{2l_4l_3l_2l_3} + \lambda_{2l_1l_2l_2l_4} \lambda_{2l_4l_3l_1l_3} + \lambda_{2l_1l_2l_3l_4} \lambda_{2l_4l_3l_1l_2} + \\
& \lambda_{2l_1l_3l_1l_4} \lambda_{2l_4l_2l_2l_3} + \lambda_{2l_1l_3l_2l_4} \lambda_{2l_4l_2l_1l_3} + \lambda_{2l_1l_3l_3l_4} \lambda_{2l_4l_2l_1l_2} + \\
& \lambda_{2l_2l_3l_1l_4} \lambda_{2l_4l_1l_2l_3} + \lambda_{2l_2l_3l_2l_4} \lambda_{2l_4l_1l_1l_3} + \lambda_{2l_2l_3l_3l_4} \lambda_{2l_4l_1l_1l_2} = 0
\end{aligned} \tag{35}$$

which gives for example

$$\begin{aligned}
& \lambda_{2121l} \lambda_{2l323} + \lambda_{2122l} \lambda_{2l313} + \lambda_{2123l} \lambda_{2l312} + \\
& \lambda_{2131l} \lambda_{2l223} + \lambda_{2132l} \lambda_{2l213} + \lambda_{2133l} \lambda_{2l212} + \\
& \lambda_{2231l} \lambda_{2l123} + \lambda_{2232l} \lambda_{2l113} + \lambda_{2233l} \lambda_{2l112} = 0
\end{aligned} \tag{36}$$

conjecture

$$\Lambda_{223} = \Gamma (L_4^3 (\lambda_2 \wedge \lambda_2)) = 0 \tag{37}$$

- two particle operators

$$\{ \lambda_{2l_1l_2k_3k_4}^2 - \lambda_{2l_1l_2k_3k_4} \} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} = 0. \tag{38}$$

A. Relation to First Quantization

Noting that

$$\begin{aligned}\lambda_2 &= \Gamma\left(\hat{\lambda}_2\right); \hat{\lambda}_2 \in \mathcal{B}\left(\mathcal{H}^2\right) \\ \hat{\lambda}_2 &= \Gamma^{-1}\left(\lambda_2\right)\end{aligned}\tag{39}$$

one obtains

$$\begin{aligned}(\lambda_2)^2 &= 36 \sum_{\substack{1 \leq l_1 < l_2 < k_1 < k_2 \leq r \\ 1 \leq l_3 < l_4 < k_3 < k_4 \leq r \\ \sigma, \eta \in S_4}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)\eta(l_4)} \lambda_{2\sigma(k_1)\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} \\ &+ 9 \sum_{\substack{1 \leq l_1 < l_2 < k_2 \leq r \\ 1 \leq l_3 < k_3 < k_4 \leq r \\ 1 \leq l_4 \leq r}} \sum_{\sigma, \eta \in S_3} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)l_4} \lambda_{2l_4\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} \\ &+ \frac{1}{4} \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq k_3 < k_4 \leq r \\ 1 \leq l_3, l_4 \leq r}} \sum_{\sigma, \eta \in S_2} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)l_3l_4} \lambda_{2l_3l_4\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \\ &= \Gamma\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right) + \Gamma\left(L_4^3\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right)\right) + \Gamma\left(\hat{\lambda}_2^2\right) = \Gamma\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2 + L_4^3\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right) + \hat{\lambda}_2^2\right)\end{aligned}\tag{40}$$

thus

$$\begin{aligned}(\lambda_2)^2 - \lambda_2 &= \Gamma\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2 + L_4^3\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right) + \hat{\lambda}_2^2 - \hat{\lambda}_2\right) = 0 \\ &\Downarrow \\ \hat{\lambda}_2^2 - \hat{\lambda}_2 &= 0 \ \& \ L_4^3\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right) = 0 \ \& \ \hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0.\end{aligned}\tag{41}$$

The condition

$$\hat{\lambda}_2^2 - \hat{\lambda}_2 = 0\tag{42}$$

shows that $\hat{\lambda}_2$ has eigenvalues $\{\alpha_P \mid 1 \leq P \leq \binom{r}{2}\} \equiv \{0, 1\}$ and the condition $\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0$ gives

$$\text{Tr}\left\{\hat{\lambda}_2 \wedge \hat{\lambda}_2\right\} = \sum_{1 \leq P < Q \leq \binom{r}{2}} \alpha_P \alpha_Q = 0\tag{43}$$

thus

$$\text{rank}\left\{\hat{\lambda}_2\right\} = 1\tag{44}$$

i.e.

$$\hat{\lambda}_2 = |g\rangle \langle g|.\tag{45}$$

Now again using $\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0$ gives

$$|g\rangle \langle g| \wedge |g\rangle \langle g| = |g \wedge g\rangle \langle g \wedge g| = 0, \quad (46)$$

thus

$$|g \wedge g\rangle = 0. \quad (47)$$

Utilizing the canonical expansion of $|g\rangle$ gives

$$\sum_{1 \leq l_1 < l_2 \leq s} c_{l_1} c_{l_2} |\psi_{l_1} \psi_{l_1+s} \psi_{l_2} \psi_{l_2+s}\rangle = 0 \quad (48)$$

which shows that

$$\text{rank } \{|g\rangle\} = 1 \quad (49)$$

i.e.

$$|g\rangle = |\psi_1 \psi_{1+s}\rangle. \quad (50)$$

III. IDEMPOTENCY OF $\lambda_1 + \lambda_2$

The general operator is given by

$$\lambda_1 + \lambda_2 = \sum_{1 \leq l \leq r} \alpha_l n_l + \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3}. \quad (51)$$

[examples of projectors are

$$\begin{aligned} & a_1^\dagger a_1 - a_1^\dagger a_2^\dagger a_2 a_1 \\ & \left(a_1^\dagger a_1 - a_1^\dagger a_2^\dagger a_2 a_1 \right)^2 = a_1^\dagger a_1 - a_1^\dagger a_1 a_1^\dagger a_2^\dagger a_2 a_1 - a_1^\dagger a_2^\dagger a_2 a_1 a_1^\dagger a_1 + a_1^\dagger a_2^\dagger a_2 a_1 \\ & = a_1^\dagger a_1 - a_1^\dagger \left(1 - a_1^\dagger a_1 \right) a_2^\dagger a_2 a_1 - a_1^\dagger a_2^\dagger a_2 \left(1 - a_1^\dagger a_1 \right) a_1 + a_1^\dagger a_2^\dagger a_2 a_1 \\ & = a_1^\dagger a_1 - a_1^\dagger a_2^\dagger a_2 a_1 - a_1^\dagger a_2^\dagger a_2 a_1 + a_1^\dagger a_2^\dagger a_2 a_1 = a_1^\dagger a_1 - a_1^\dagger a_2^\dagger a_2 a_1 \end{aligned} \quad (52)$$

$$\begin{aligned} \lambda_1^2 &= a_1^\dagger a_1 \\ \lambda_1^2 - \lambda_1 &= 0 \\ \lambda_2 &= -a_1^\dagger a_2^\dagger a_2 a_1 \\ \lambda_2^2 &= a_1^\dagger a_2^\dagger a_2 a_1 \\ \lambda_2^2 - \lambda_2 &= 2a_1^\dagger a_2^\dagger a_2 a_1 \end{aligned} \quad (53)$$

$$\begin{aligned}
& a_1^\dagger a_1 + a_2^\dagger a_2 - a_1^\dagger a_2^\dagger a_2 a_1 \\
& \left(a_1^\dagger a_1 + a_2^\dagger a_2 - a_1^\dagger a_2^\dagger a_2 a_1 \right)^2 = \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right)^2 - \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) a_1^\dagger a_2^\dagger a_2 a_1 - a_1^\dagger a_2^\dagger a_2 a_1 \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) + a_1^\dagger a_2^\dagger a_2 a_1 \\
& = a_1^\dagger a_1 + a_2^\dagger a_2 + 2a_1^\dagger a_2^\dagger a_2 a_1 - a_1^\dagger \left(1 - a_1^\dagger a_1 \right) a_2^\dagger a_2 a_1 - a_1^\dagger \left(1 - a_2^\dagger a_2 \right) a_2 a_1 - a_1^\dagger a_2^\dagger a_2 \left(1 - a_1^\dagger a_1 \right) a_1 - a_1^\dagger a_2^\dagger \left(1 - a_2^\dagger a_2 \right) a_2 \\
& = a_1^\dagger a_1 + a_2^\dagger a_2 + 2a_1^\dagger a_2^\dagger a_2 a_1 - 4a_1^\dagger a_2^\dagger a_2 a_1 + a_1^\dagger a_2^\dagger a_2 a_1 = a_1^\dagger a_1 + a_2^\dagger a_2 - a_1^\dagger a_2^\dagger a_2 a_1 \quad (54)
\end{aligned}$$

$$\begin{aligned}
\lambda_1^2 &= a_1^\dagger a_1 + a_2^\dagger a_2 + 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1^2 - \lambda_1 &= 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2 &= -a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2^2 &= a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2^2 - \lambda_2 &= 2a_1^\dagger a_2^\dagger a_2 a_1 \quad (55)
\end{aligned}$$

$$\begin{aligned}
(\lambda_1 + \lambda_2)^2 - \lambda_1 - \lambda_2 &= \lambda_1^2 - \lambda_1 + \lambda_2^2 - \lambda_2 + \lambda_1 \lambda_2 + \lambda_2 \lambda_1 \\
&= 2a_1^\dagger a_2^\dagger a_2 a_1 + 2a_1^\dagger a_2^\dagger a_2 a_1 - 4a_1^\dagger a_2^\dagger a_2 a_1 \quad (56)
\end{aligned}$$

The idempotency condition leads to

$$(\lambda_1 + \lambda_2)^2 - \lambda_1 - \lambda_2 = \lambda_1^2 - \lambda_1 + \lambda_2^2 - \lambda_2 + \lambda_1 \lambda_2 + \lambda_2 \lambda_1 = 0. \quad (57)$$

The only new term is

$$\lambda_1 \lambda_2 + \lambda_2 \lambda_1, \quad (58)$$

[for example

1.

$$\begin{aligned}
\lambda_1 &= a_1^\dagger a_1; \quad \lambda_2 = -a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1 \lambda_2 &= -a_1^\dagger a_1 a_1^\dagger a_2^\dagger a_2 a_1 = -a_1^\dagger \left(1 - a_1^\dagger a_1 \right) a_2^\dagger a_2 a_1 = -a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2 \lambda_1 &= -a_1^\dagger a_2^\dagger a_2 a_1 a_1^\dagger a_1 = -a_1^\dagger a_2^\dagger a_2 \left(1 - a_1^\dagger a_1 \right) a_1 = -a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1 \lambda_2 + \lambda_2 \lambda_1 &= -2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1^2 - \lambda_1 &= 0 \\
\lambda_2^2 - \lambda_2 &= 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1^2 - \lambda_1 + \lambda_2^2 - \lambda_2 + \lambda_1 \lambda_2 + \lambda_2 \lambda_1 &= 0 + 2a_1^\dagger a_2^\dagger a_2 a_1 - 2a_1^\dagger a_2^\dagger a_2 a_1 = 0 \quad (59)
\end{aligned}$$

2.

$$\begin{aligned}
\lambda_1 &= a_1^\dagger a_1 + a_2^\dagger a_2; \quad \lambda_2 = -a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1 \lambda_2 &= -\left(a_1^\dagger a_1 + a_2^\dagger a_2\right) a_1^\dagger a_2^\dagger a_2 a_1 = -a_1^\dagger \left(1 - a_1^\dagger a_1\right) a_2^\dagger a_2 a_1 - a_2^\dagger \left(1 - a_2^\dagger a_2\right) a_1^\dagger a_1 a_2 = -2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2 \lambda_1 &= -2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1 \lambda_2 + \lambda_2 \lambda_1 &= -4a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1^2 - \lambda_1 &= \left(a_1^\dagger a_1 + a_2^\dagger a_2\right)^2 - \left(a_1^\dagger a_1 + a_2^\dagger a_2\right) = 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2^2 - \lambda_2 &= a_1^\dagger a_2^\dagger a_2 a_1 - -a_1^\dagger a_2^\dagger a_2 a_1 = 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1^2 - \lambda_1 + \lambda_2^2 - \lambda_2 + \lambda_1 \lambda_2 + \lambda_2 \lambda_1 &= 0 + 2a_1^\dagger a_2^\dagger a_2 a_1 + 2a_1^\dagger a_2^\dagger a_2 a_1 - 4a_1^\dagger a_2^\dagger a_2 a_1 = 0]
\end{aligned} \tag{60}$$

which we now evaluate and obtain

$$\begin{aligned}
\lambda_1 \lambda_2 &= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_l^\dagger a_l a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} = \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_l^\dagger \left(\delta_{ll_1} - a_{l_1}^\dagger a_l\right) a_{l_2}^\dagger a_{l_4} a_{l_3} \\
&= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l \left\{ a_l^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \delta_{ll_1} - a_l^\dagger a_{l_1}^\dagger \left(a_l a_{l_2}^\dagger\right) a_{l_4} a_{l_3} \right\} \\
&= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l \left\{ a_l^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \delta_{ll_1} - a_l^\dagger a_{l_1}^\dagger \left(\delta_{ll_2} - a_{l_2}^\dagger a_l\right) a_{l_4} a_{l_3} \right\} \\
&= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l \left\{ a_l^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \delta_{ll_1} - a_l^\dagger a_{l_1}^\dagger a_{l_4} a_{l_3} \delta_{ll_2} + a_l^\dagger a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_l a_{l_4} a_{l_3} \right\} \\
&= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_l^\dagger a_{l_1}^\dagger a_{l_2}^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_1} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \\
&\quad + \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \\
&= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \{\alpha_{l_1} + \alpha_{l_2}\} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3}, \tag{61}
\end{aligned}$$

which gives

$$\lambda_1 \lambda_2 = \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{2} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_1} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3}. \quad (62)$$

The adjoint is

$$\begin{aligned} \lambda_2 \lambda_1 &= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} \left(a_{l_3} a_l^\dagger \right) a_l = \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} \left(\delta_{ll_3} - a_l^\dagger a_{l_3} \right) a_l \\ &= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l \left\{ a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_l \delta_{ll_3} - a_{l_1}^\dagger a_{l_2}^\dagger \left(a_{l_4} a_l^\dagger \right) a_{l_3} a_l \right\} \\ &= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l \left\{ a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_l \delta_{ll_3} - a_{l_1}^\dagger a_{l_2}^\dagger a_{l_3} a_l \delta_{ll_4} + a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_{l_4} a_{l_3} a_l \right\} \\ &= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_{l_4} a_{l_3} a_l + \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_3} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \\ &\quad + \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_4} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \\ &= \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_3} + \alpha_{l_4}) a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \quad (63) \end{aligned}$$

$$\lambda_2 \lambda_1 = \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{2} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_3} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \quad (64)$$

$$\lambda_1 \lambda_2 = \frac{1}{4} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{2} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_{l_1} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \quad (65)$$

Hence

$$\lambda_1 \lambda_2 + \lambda_2 \lambda_1 = \frac{1}{2} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \frac{1}{2} \sum_{\substack{1 \leq l_1, l_2 \leq r \\ 1 \leq l_3, l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_1} + \alpha_{l_3}) a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \quad (66)$$

$$= 2 \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} + \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_1} + \alpha_{l_2} + \alpha_{l_3} + \alpha_{l_4}) a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \quad (67)$$

and we define

$$\Lambda_{123} = 2 \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} \quad (68)$$

$$\Lambda_{122} = \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_1} + \alpha_{l_2} + \alpha_{l_3} + \alpha_{l_4}) a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} \quad (69)$$

and the idempotency condition becomes

$$(\lambda_1 + \lambda_2)^2 - (\lambda_1 + \lambda_2) = \Lambda_{111} + \Lambda_{112} + \Lambda_{224} + \Lambda_{223} + \Lambda_{222} + \Lambda_{122} + \Lambda_{123} = 0.$$

Collecting together the operators of the same order one obtains by considering linear dependence

$$\begin{aligned} \Lambda_{111} &= \sum_{1 \leq l \leq r} \alpha_l (\alpha_l - 1) a_l^\dagger a_l = 0 \Leftrightarrow \alpha_l = 1 \text{ or } \alpha_l = 0 \\ \Lambda_{112} + \Lambda_{222} + \Lambda_{122} &= 2 \sum_{1 \leq l_1 < l_2 \leq r} \alpha_{l_1} \alpha_{l_2} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} + \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq k_3 < k_4 \leq r}} \{ \lambda_{2l_1 l_2 k_3 k_4}^2 - \lambda_{2l_1 l_2 k_3 k_4} \} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} \\ &\quad + \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r}} \lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_1} + \alpha_{l_2} + \alpha_{l_3} + \alpha_{l_4}) a_{l_1}^\dagger a_{l_2}^\dagger a_{l_4} a_{l_3} = 0 \\ \Lambda_{223} + \Lambda_{123} &= 9 \sum_{\substack{1 \leq l_1 < l_2 < k_2 \leq r \\ 1 \leq l_3 < k_3 < k_4 \leq r}} \sum_{\substack{\sigma, \eta \in S_3 \\ 1 \leq l_4 \leq r}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)l_4} \lambda_{2l_4\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_2}^\dagger a_{l_3} a_{k_4} a_{k_3} \\ &\quad + \sum_{\substack{1 \leq l_1 < l_2 \leq r \\ 1 \leq l_3 < l_4 \leq r \\ 1 \leq l \leq r}} \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} = 0 \\ \Lambda_{224} &= 36 \sum_{\substack{1 \leq l_1 < l_2 < k_1 < k_2 \leq r \\ 1 \leq l_3 < l_4 < k_3 < k_4 \leq r \\ \sigma, \eta \in S_4}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)\eta(l_4)} \lambda_{2\sigma(k_1)\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} \\ &= 0. \end{aligned} \quad (70a)$$

One can then use linear independency to obtain

$$\begin{aligned}
& \alpha_l (\alpha_l - 1) a_l^\dagger a_l = 0, \\
& \{2\alpha_{l_1} \alpha_{l_2} + (\lambda_{2l_1 l_2 l_1 l_2}^2 - \lambda_{2l_1 l_2 l_1 l_2}) + 2\lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_1} + \alpha_{l_2})\} a_{l_1}^\dagger a_{l_2}^\dagger a_{l_2} a_{l_1} = 0, \\
& \{\lambda_{2l_1 l_2 k_3 k_4}^2 - \lambda_{2l_1 l_2 k_3 k_4} + \lambda_{2l_1 l_2 l_3 l_4} (\alpha_{l_1} + \alpha_{l_2} + \alpha_{l_3} + \alpha_{l_4})\} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_4} a_{k_3} = 0, l_1 l_2 \neq k_3 k_4 \quad (71a)
\end{aligned}$$

$$\sum_{\substack{\sigma, \eta \in S_3 \\ 1 \leq l_4 \leq r}} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)l_4} \lambda_{2l_4\sigma(l)\eta(l_3)\eta(l_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} \quad (72)$$

$$+ \lambda_{2l_1 l_2 l_3 l_4} \alpha_l a_{l_1}^\dagger a_{l_2}^\dagger a_l^\dagger a_l a_{l_4} a_{l_3} = 0 \quad (73)$$

$$\sum_{\sigma, \eta \in S_3} (-1)^{\pi(\sigma)} (-1)^{\pi(\eta)} \lambda_{2\sigma(l_1)\sigma(l_2)\eta(l_3)\eta(l_4)} \lambda_{2\sigma(k_1)\sigma(k_2)\eta(k_3)\eta(k_4)} a_{l_1}^\dagger a_{l_2}^\dagger a_{k_1}^\dagger a_{k_2}^\dagger a_{l_4} a_{l_3} a_{k_4} a_{k_3} = 0 \quad (74)$$

A. Relation to First Quantization

- First one can note that

$$(\lambda_1 + \lambda_2)^2 = \lambda_1 + \lambda_2 \Rightarrow \lambda_1 + \lambda_2 \geq 0, \quad (75)$$

which shows

$$\hat{\lambda}_1 \geq 0 \Leftrightarrow \lambda_1 \geq 0 \quad (76)$$

and

$$\lambda_2 = \lambda_2^\dagger. \quad (77)$$

Then noting that

$$\begin{aligned}
\lambda_1 + \lambda_2 &= \Gamma(\hat{\lambda}_1 + \hat{\lambda}_2); \hat{\lambda}_1 \in \mathcal{B}(\mathcal{H}^1), \hat{\lambda}_2 \in \mathcal{B}(\mathcal{H}^2) \\
\hat{\lambda}_1 + \hat{\lambda}_2 &= \Gamma^{-1}(\lambda_1 + \lambda_2)
\end{aligned} \quad (78)$$

one obtains

$$\begin{aligned}
& (\lambda_1)^2 + \lambda_1 \lambda_2 + \lambda_2 \lambda_1 + (\lambda_2)^2 = \\
& \Gamma(\hat{\lambda}_1^2) + \Gamma(\hat{\lambda}_1 \wedge \hat{\lambda}_1) + 2\Gamma(\hat{\lambda}_1 \wedge \hat{\lambda}_2) + 2\Gamma(L_3^2(\hat{\lambda}_1 \wedge \hat{\lambda}_2)) + \Gamma(\hat{\lambda}_2 \wedge \hat{\lambda}_2) + \Gamma(L_3^4(\hat{\lambda}_2 \wedge \hat{\lambda}_2)) + \Gamma(\hat{\lambda}_2^2) \\
& \Gamma(\hat{\lambda}_1^2 + \hat{\lambda}_1 \wedge \hat{\lambda}_1 + 2\hat{\lambda}_1 \wedge \hat{\lambda}_2 + 2L_3^2(\hat{\lambda}_1 \wedge \hat{\lambda}_2) + \hat{\lambda}_2 \wedge \hat{\lambda}_2 + L_3^4(\hat{\lambda}_2 \wedge \hat{\lambda}_2) + \hat{\lambda}_2^2) \quad (79)
\end{aligned}$$

[for example

$$\begin{aligned}
\lambda_1 &= a_1^\dagger a_1 + a_2^\dagger a_2 = \Gamma(|\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2|); \lambda_2 = -a_1^\dagger a_2^\dagger a_2 a_1 = -\Gamma(|\psi_1\psi_2\rangle\langle\psi_1\psi_2|) \\
\lambda_1\lambda_2 &= -\left(a_1^\dagger a_1 + a_2^\dagger a_2\right) a_1^\dagger a_2^\dagger a_2 a_1 = -a_1^\dagger \left(1 - a_1^\dagger a_1\right) a_2^\dagger a_2 - a_2^\dagger \left(1 - a_2^\dagger a_2\right) a_1^\dagger a_1 a_2 = -2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2\lambda_1 &= -2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1\lambda_2 + \lambda_2\lambda_1 &= -4a_1^\dagger a_2^\dagger a_2 a_1 = -4\Gamma(|\psi_1\psi_2\rangle\langle\psi_1\psi_2|) \\
\lambda_1^2 - \lambda_1 &= \left(a_1^\dagger a_1 + a_2^\dagger a_2\right)^2 - \left(a_1^\dagger a_1 + a_2^\dagger a_2\right) = 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_2^2 - \lambda_2 &= a_1^\dagger a_2^\dagger a_2 a_1 - -a_1^\dagger a_2^\dagger a_2 a_1 = 2a_1^\dagger a_2^\dagger a_2 a_1 \\
\lambda_1^2 - \lambda_1 + \lambda_2^2 - \lambda_2 + \lambda_1\lambda_2 + \lambda_2\lambda_1 &= 0 + 2a_1^\dagger a_2^\dagger a_2 a_1 + 2a_1^\dagger a_2^\dagger a_2 a_1 - 4a_1^\dagger a_2^\dagger a_2 a_1 = 0]
\end{aligned} \tag{80}$$

hence the idempotency condition

$$(\lambda_1 + \lambda_2)^2 - (\lambda_1 + \lambda_2) = \lambda_1^2 - \lambda_1 + \lambda_2^2 - \lambda_2 + \lambda_1\lambda_2 + \lambda_2\lambda_1 = 0. \tag{81}$$

This gives

$$\begin{aligned}
&(\lambda_1 + \lambda_2)^2 - (\lambda_1 + \lambda_2) = \\
&\Gamma\left(\hat{\lambda}_1^2 + \hat{\lambda}_1 \wedge \hat{\lambda}_1 + 2\hat{\lambda}_1 \wedge \hat{\lambda}_2 + \Gamma_1^2\left(\hat{\lambda}_1\right) \hat{\lambda}_2 + \hat{\lambda}_2 \Gamma_1^2\left(\hat{\lambda}_1\right) \right. \\
&\quad \left. + \hat{\lambda}_2 \wedge \hat{\lambda}_2 + L_4^3\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right) + \hat{\lambda}_2^2 - \hat{\lambda}_1 - \hat{\lambda}_2\right) \\
&\quad \Updownarrow
\end{aligned} \tag{82}$$

$$\begin{aligned}
&\hat{\lambda}_1^2 - \hat{\lambda}_1 = 0 \\
&\hat{\lambda}_1 \wedge \hat{\lambda}_1 + \hat{\lambda}_2^2 - \hat{\lambda}_2 + \Gamma_1^2\left(\hat{\lambda}_1\right) \hat{\lambda}_2 + \hat{\lambda}_2 \Gamma_1^2\left(\hat{\lambda}_1\right) = 0 \\
&2\hat{\lambda}_1 \wedge \hat{\lambda}_2 + L_4^3\left(\hat{\lambda}_2 \wedge \hat{\lambda}_2\right) = 0 \Rightarrow \hat{\lambda}_1 \wedge \hat{\lambda}_2 = 0 \\
&\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0
\end{aligned} \tag{83}$$

$$\begin{aligned}
\lambda_1 + \lambda_2 &= \Gamma\left(\hat{\lambda}_1 + \hat{\lambda}_2\right) \\
&= \sum_j \lambda_{1jj} \left\{ |\psi_j\rangle\langle\psi_j| + \sum_{i_1} |\psi_j\psi_{i_1}\rangle\langle\psi_j\psi_{i_1}| + \sum_{i_1 i_2} |\psi_j\psi_{i_1}\psi_{i_2}\rangle\langle\psi_j\psi_{i_1}\psi_{i_2}| + \dots \right\} \\
&+ \sum_{j_1 < j_2, k_1 < k_2} \lambda_{2j_1 j_2 k_1 k_2} \left\{ |\psi_{j_1}\psi_{j_2}\rangle\langle\psi_{k_1}\psi_{k_2}| + \sum_{i_1} |\psi_{j_1}\psi_{j_2}\psi_{i_1}\rangle\langle\psi_{k_1}\psi_{k_2}\psi_{i_1}| \right. \\
&\quad \left. + \sum_{i_1 < i_2} |\psi_{j_1}\psi_{j_2}\psi_{i_1}\psi_{i_2}\rangle\langle\psi_{k_1}\psi_{k_2}\psi_{i_1}\psi_{i_2}| + \dots \right\}
\end{aligned} \tag{84}$$

- The condition

$$\hat{\lambda}_1^2 - \hat{\lambda}_1 = 0 \quad (85)$$

shows that $\hat{\lambda}_1$ has eigenvalues $\{0, 1\}$ and $\hat{\lambda}_1$ is a projector which can be expanded as

$$\hat{\lambda}_1 = \sum_{1 \leq j \leq v} |\psi_j\rangle \langle \psi_j| \quad (86)$$

for any basis that spans the range of $\hat{\lambda}_1$.

- The condition $\hat{\lambda}_1 \wedge \hat{\lambda}_2 = 0$ shows that

$$\hat{\lambda}_1 \wedge \hat{\lambda}_2 = \sum_{1 \leq j \leq v} |\psi_j\rangle \langle \psi_j| \wedge \sum_{\substack{1 \leq k < l \leq r \\ 1 \leq m < n \leq r}} \lambda_{klmn} |\psi_k \psi_l\rangle \langle \psi_m \psi_n| = \sum_{\substack{1 \leq k < l \leq r \\ 1 \leq m < n \leq r \\ 1 \leq j \leq v}} \lambda_{klmn} |\psi_j \psi_k \psi_l\rangle \langle \psi_j \psi_m \psi_n| = 0. \quad (87)$$

Given a projector $\hat{\lambda}_1$ what solutions for $\hat{\lambda}_2$ are possible for $\hat{\lambda}_1 \wedge \hat{\lambda}_2 = 0$, first we determine the subspace $\{|\psi_k \psi_l\rangle \langle \psi_m \psi_n|\}$, which is the kernel of the linear map $\sum_{1 \leq j \leq v} |\psi_j\rangle \langle \psi_j| \wedge$ i.e.

$$\sum_{1 \leq j \leq v} |\psi_j\rangle \langle \psi_j| \wedge |\psi_k \psi_l\rangle \langle \psi_m \psi_n| = 0. \quad (88)$$

This gives for $\nu = 1$

$$|\psi_1\rangle \langle \psi_1| \wedge |\psi_k \psi_l\rangle \langle \psi_m \psi_n| = 0 \Rightarrow k = 1 \text{ or } l = 1 \text{ or } m = 1 \text{ or } n = 1 \quad (89)$$

i.e. $|\psi_k \psi_l\rangle \langle \psi_m \psi_n|$ must be of the form

$$\{|\psi_1 \psi_l\rangle \langle \psi_m \psi_n|, |\psi_k \psi_l\rangle \langle \psi_1 \psi_n|, |\psi_1 \psi_l\rangle \langle \psi_1 \psi_n|\}. \quad (90)$$

For $\nu = 2$ the operators $\{|\psi_k \psi_l\rangle \langle \psi_m \psi_n|\}$ must be "orthogonal" to both $|\psi_1\rangle \langle \psi_1|$ and $|\psi_2\rangle \langle \psi_2|$ and must be of the form

$$|\psi_1 \psi_l\rangle \langle \psi_2 \psi_n|, |\psi_k \psi_l\rangle \langle \psi_1 \psi_2|, |\psi_2 \psi_l\rangle \langle \psi_1 \psi_n|, |\psi_1 \psi_2\rangle \langle \psi_m \psi_n|. \quad (91)$$

For $\nu = 3$ the operators $\{|\psi_k \psi_l\rangle \langle \psi_m \psi_n|\}$ must be "orthogonal" to $|\psi_1\rangle \langle \psi_1|$, $|\psi_2\rangle \langle \psi_2|$ and $|\psi_3\rangle \langle \psi_3|$ thus they must be of the form

$$|\psi_1 \psi_3\rangle \langle \psi_2 \psi_n|, |\psi_1 \psi_l\rangle \langle \psi_2 \psi_3|, |\psi_k \psi_3\rangle \langle \psi_1 \psi_2|, |\psi_2 \psi_3\rangle \langle \psi_1 \psi_n|, |\psi_2 \psi_l\rangle \langle \psi_1 \psi_3|, |\psi_1 \psi_2\rangle \langle \psi_3 \psi_n|. \quad (92)$$

For $\nu = 4$ the operators $\{|\psi_k\psi_l\rangle\langle\psi_m\psi_n|\}$ must be "orthogonal" to $|\psi_1\rangle\langle\psi_1|$, $|\psi_2\rangle\langle\psi_2|$, $|\psi_3\rangle\langle\psi_3|$ and $|\psi_4\rangle\langle\psi_4|$ and must be of the form

$$|\psi_1\psi_3\rangle\langle\psi_2\psi_4|, |\psi_1\psi_4\rangle\langle\psi_2\psi_3|, |\psi_3\psi_4\rangle\langle\psi_1\psi_2|, |\psi_2\psi_3\rangle\langle\psi_1\psi_4|, |\psi_2\psi_4\rangle\langle\psi_1\psi_3|, |\psi_1\psi_2\rangle\langle\psi_3\psi_4|. \quad (93)$$

The operator $\hat{\lambda}_2$ is self adjoint thus one needs to consider basis elements

$$\left\{ \frac{1}{\sqrt{2}} (|\psi_k\psi_l\rangle\langle\psi_m\psi_n| + |\psi_m\psi_n\rangle\langle\psi_k\psi_l|), \frac{i}{\sqrt{2}} (|\psi_k\psi_l\rangle\langle\psi_m\psi_n| - |\psi_m\psi_n\rangle\langle\psi_k\psi_l|) \right\} \quad (94)$$

which in order to be orthogonal to $|\psi_1\rangle\langle\psi_1|$ and $|\psi_2\rangle\langle\psi_2|$ must be of the form

$$\begin{aligned} & \frac{1}{\sqrt{2}} (|\psi_1\psi_l\rangle\langle\psi_2\psi_n| + |\psi_2\psi_l\rangle\langle\psi_1\psi_n|), \frac{1}{\sqrt{2}} (|\psi_k\psi_l\rangle\langle\psi_1\psi_2| + |\psi_1\psi_2\rangle\langle\psi_k\psi_l|) \\ & \frac{i}{\sqrt{2}} (|\psi_1\psi_l\rangle\langle\psi_2\psi_n| - |\psi_2\psi_l\rangle\langle\psi_1\psi_n|), \frac{i}{\sqrt{2}} (|\psi_k\psi_l\rangle\langle\psi_1\psi_2| - |\psi_1\psi_2\rangle\langle\psi_k\psi_l|), \end{aligned} \quad (95)$$

while in order it be orthogonal to $|\psi_1\rangle\langle\psi_1|$, $|\psi_2\rangle\langle\psi_2|$ and $|\psi_3\rangle\langle\psi_3|$ they must be of the form

$$\begin{aligned} & \frac{1}{\sqrt{2}} (|\psi_1\psi_3\rangle\langle\psi_2\psi_n| + |\psi_2\psi_n\rangle\langle\psi_1\psi_3|), \frac{1}{\sqrt{2}} (|\psi_3\psi_l\rangle\langle\psi_1\psi_2| + |\psi_1\psi_2\rangle\langle\psi_3\psi_l|) \\ & \frac{1}{\sqrt{2}} (|\psi_2\psi_3\rangle\langle\psi_1\psi_n| + |\psi_1\psi_n\rangle\langle\psi_2\psi_3|), \\ & \frac{i}{\sqrt{2}} (|\psi_1\psi_3\rangle\langle\psi_2\psi_n| - |\psi_2\psi_n\rangle\langle\psi_1\psi_3|), \frac{i}{\sqrt{2}} (|\psi_3\psi_l\rangle\langle\psi_1\psi_2| - |\psi_1\psi_2\rangle\langle\psi_3\psi_l|) \\ & \frac{i}{\sqrt{2}} (|\psi_2\psi_3\rangle\langle\psi_1\psi_n| - |\psi_1\psi_n\rangle\langle\psi_2\psi_3|), \end{aligned} \quad (96)$$

while in order be orthogonal to $|\psi_1\rangle\langle\psi_1|$, $|\psi_2\rangle\langle\psi_2|$, $|\psi_3\rangle\langle\psi_3|$ and $|\psi_4\rangle\langle\psi_4|$ it must be of the form

$$\begin{aligned} & \frac{1}{\sqrt{2}} (|\psi_1\psi_3\rangle\langle\psi_2\psi_4| + |\psi_2\psi_4\rangle\langle\psi_1\psi_3|), \frac{1}{\sqrt{2}} (|\psi_1\psi_2\rangle\langle\psi_3\psi_4| + |\psi_3\psi_4\rangle\langle\psi_1\psi_2|) \\ & \frac{1}{\sqrt{2}} (|\psi_1\psi_4\rangle\langle\psi_2\psi_3| + |\psi_2\psi_3\rangle\langle\psi_1\psi_4|), \\ & |\psi_1\psi_2\rangle\langle\psi_1\psi_2|, |\psi_1\psi_3\rangle\langle\psi_1\psi_3|, |\psi_1\psi_4\rangle\langle\psi_1\psi_4|, |\psi_2\psi_3\rangle\langle\psi_2\psi_3|, |\psi_2\psi_4\rangle\langle\psi_2\psi_4|, |\psi_3\psi_4\rangle\langle\psi_3\psi_4|, \\ & \frac{i}{\sqrt{2}} (|\psi_1\psi_3\rangle\langle\psi_2\psi_4| - |\psi_2\psi_4\rangle\langle\psi_1\psi_3|), \frac{i}{\sqrt{2}} (|\psi_3\psi_4\rangle\langle\psi_1\psi_2| - |\psi_1\psi_2\rangle\langle\psi_3\psi_4|) \\ & \frac{i}{\sqrt{2}} (|\psi_2\psi_3\rangle\langle\psi_1\psi_4| - |\psi_1\psi_4\rangle\langle\psi_2\psi_3|). \end{aligned} \quad (97)$$

- We consider the condition

$$\hat{\lambda}_1 \wedge \hat{\lambda}_2 = 0 \quad (98)$$

- 1. Let

$$\hat{\lambda}_1 = |\psi_1\rangle \langle \psi_1| \quad (99)$$

then $\lambda_2 \equiv \lambda$

$$\hat{\lambda}_2 = \sum_{\substack{1 \leq i < r \\ 1 \leq j < k \leq r}} \{ \lambda_{1ijk} |\psi_1 \psi_i\rangle \langle \psi_j \psi_k| + \lambda_{jk1i} |\psi_j \psi_k\rangle \langle \psi_1 \psi_i| \} + \sum_{1 < i < r} \lambda_{1i1j} |\psi_1 \psi_i\rangle \langle \psi_1 \psi_j|. \quad (100)$$

- 2. Let

$$\hat{\lambda}_1 = |\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| \quad (101)$$

then

$$\begin{aligned} \hat{\lambda}_2 = & \sum \lambda_{12ik} |\psi_1 \psi_2\rangle \langle \psi_i \psi_k| + \sum \lambda_{1i2k} |\psi_1 \psi_i\rangle \langle \psi_2 \psi_k| + \sum \lambda_{2i1k} |\psi_2 \psi_i\rangle \langle \psi_1 \psi_k| + \sum \lambda_{jk1i} |\psi_i \psi_k\rangle \langle \psi_1 \psi_2| \\ & + \sum_{1 < i < r} \lambda_{121i} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_i| + \sum_{1 < i < r} \lambda_{121i} |\psi_1 \psi_i\rangle \langle \psi_1 \psi_2| + \sum_{1 < i < r} \lambda_{122i} |\psi_1 \psi_2\rangle \langle \psi_2 \psi_i| + \sum_{1 < i < r} \lambda_{1i12} |\psi_2 \psi_i\rangle \langle \psi_1 \psi_2| \\ & + \lambda_{1212} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_2|. \end{aligned} \quad (102)$$

The condition $\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0$ gives

$$\hat{\lambda}_2 \wedge \hat{\lambda}_2 = \sum \lambda_{12ik} |\psi_1 \psi_2\rangle \langle \psi_i \psi_k| \wedge \sum \lambda_{jk1i} |\psi_i \psi_k\rangle \langle \psi_1 \psi_2| + \sum \lambda_{jk1i} |\psi_i \psi_k\rangle \langle \psi_1 \psi_2| \wedge \sum \lambda_{12ik} |\psi_1 \psi_2\rangle \langle \psi_i \psi_k|$$

- 3. Let

$$\hat{\lambda}_1 = |\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| \quad (103)$$

then

$$\begin{aligned} \hat{\lambda}_2 = & \sum_{1 \leq i \leq r} \{ \lambda_{123i} |\psi_1 \psi_2\rangle \langle \psi_3 \psi_i| + \lambda_{132i} |\psi_1 \psi_3\rangle \langle \psi_2 \psi_i| + \lambda_{231i} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_i| + \\ & \lambda_{3i12} |\psi_3 \psi_i\rangle \langle \psi_1 \psi_2| + \lambda_{2i13} |\psi_2 \psi_i\rangle \langle \psi_1 \psi_3| + \lambda_{1i23} |\psi_1 \psi_i\rangle \langle \psi_2 \psi_3| \} \\ & + \lambda_{1213} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_3| + \lambda_{1312} |\psi_1 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{1223} |\psi_1 \psi_2\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{2312} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_3| + \lambda_{1323} |\psi_1 \psi_3\rangle \langle \psi_2 \psi_3|. \end{aligned} \quad (104)$$

In particular note that in this case $\hat{\lambda}_2$ has no diagonal elements. The condition $\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0$ gives

$$\begin{aligned} \hat{\lambda}_2 \wedge \hat{\lambda}_2 = & \sum_{1 \leq i \leq r} \{ |\lambda_{123i}|^2 |\psi_1 \psi_2 \psi_3 \psi_i\rangle \langle \psi_3 \psi_i \psi_1 \psi_2| + |\lambda_{132i}|^2 |\psi_1 \psi_3 \psi_2 \psi_i\rangle \langle \psi_2 \psi_i \psi_1 \psi_3| + |\lambda_{231i}|^2 |\psi_2 \psi_3 \psi_1 \psi_i\rangle \langle \psi_1 \psi_i \psi_2 \psi_3| \\ & + \lambda_{123i} \lambda_{132i} |\psi_1 \psi_2 \psi_3 \psi_i\rangle \langle \psi_3 \psi_i \psi_1 \psi_2| + \lambda_{132i} \lambda_{123i} |\psi_1 \psi_3 \psi_2 \psi_i\rangle \langle \psi_2 \psi_i \psi_1 \psi_3| + \lambda_{231i} \lambda_{132i} |\psi_2 \psi_3 \psi_1 \psi_i\rangle \langle \psi_1 \psi_i \psi_2 \psi_3| \\ & + \lambda_{123i} \lambda_{231i} |\psi_1 \psi_2 \psi_3 \psi_i\rangle \langle \psi_3 \psi_i \psi_1 \psi_2| + \lambda_{132i} \lambda_{231i} |\psi_1 \psi_3 \psi_2 \psi_i\rangle \langle \psi_2 \psi_i \psi_1 \psi_3| + \lambda_{231i} \lambda_{123i} |\psi_2 \psi_3 \psi_1 \psi_i\rangle \langle \psi_1 \psi_i \psi_2 \psi_3| \} \end{aligned} \quad (105)$$

Thus

$$\lambda_{1234} = \lambda_{1324} = \lambda_{1423} = \lambda_{3412} = \lambda_{2413} = \lambda_{2314} = 0 \quad (106)$$

giving

$$\begin{aligned} \hat{\lambda}_2 = & \lambda_{1213} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_3| + \lambda_{1312} |\psi_1 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{1223} |\psi_1 \psi_2\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{2312} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_3| + \lambda_{1323} |\psi_1 \psi_3\rangle \langle \psi_2 \psi_3|. \end{aligned} \quad (107)$$

Now we consider we consider the condition

$$\hat{\lambda}_1 \wedge \hat{\lambda}_1 + \hat{\lambda}_2^2 - \hat{\lambda}_2 + \Gamma_1^2 (\hat{\lambda}_1) \hat{\lambda}_2 + \hat{\lambda}_2 \Gamma_1^2 (\hat{\lambda}_1) = 0 \quad (108)$$

this gives

$$\begin{aligned} & (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3|) \wedge (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3|) + (|\lambda_{1213}|^2 + |\lambda_{1223}|^2) |\psi_1 \psi_2\rangle \langle \psi_1 \psi_2| \\ & + (|\lambda_{1312}|^2 + |\lambda_{1323}|^2) |\psi_1 \psi_3\rangle \langle \psi_1 \psi_3| + (|\lambda_{2312}|^2 + |\lambda_{2323}|^2) |\psi_2 \psi_3\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{1213} \lambda_{1323} |\psi_1 \psi_2\rangle \langle \psi_2 \psi_3| + \lambda_{1223} \lambda_{2313} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_3| + \lambda_{1312} \lambda_{1223} |\psi_1 \psi_3\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{2313} \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{1323} \lambda_{2312} |\psi_1 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{2313} \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_3| + \\ & (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3|) \wedge (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| + \dots) \hat{\lambda}_2 \\ & \hat{\lambda}_2 (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3|) \wedge (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| + \dots) - \hat{\lambda}_2 = 0 \end{aligned} \quad (109)$$

and hence

$$\begin{aligned} \hat{\lambda}_1 \wedge \hat{\lambda}_1 - 2 & (|\psi_1 \psi_2\rangle \langle \psi_1 \psi_2| + |\psi_1 \psi_3\rangle \langle \psi_1 \psi_3| + |\psi_2 \psi_3\rangle \langle \psi_2 \psi_3|) + (|\lambda_{1213}|^2 + |\lambda_{1223}|^2) |\psi_1 \psi_2\rangle \langle \psi_1 \psi_2| \\ & + (|\lambda_{1312}|^2 + |\lambda_{1323}|^2) |\psi_1 \psi_3\rangle \langle \psi_1 \psi_3| + (|\lambda_{2312}|^2 + |\lambda_{2323}|^2) |\psi_2 \psi_3\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{1213} \lambda_{1323} |\psi_1 \psi_2\rangle \langle \psi_2 \psi_3| + \lambda_{1223} \lambda_{2313} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_3| + \lambda_{1312} \lambda_{1223} |\psi_1 \psi_3\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{2313} \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{1323} \lambda_{2312} |\psi_1 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{2313} \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_3| + \\ & 4\hat{\lambda}_2 - \hat{\lambda}_2 = 0 \end{aligned} \quad (110)$$

leading to

$$\begin{aligned} \hat{\lambda}_1 \wedge \hat{\lambda}_1 - & (2 + |\lambda_{1213}|^2 + |\lambda_{1223}|^2) |\psi_1 \psi_2\rangle \langle \psi_1 \psi_2| + (2 + |\lambda_{1312}|^2 + |\lambda_{1323}|^2) |\psi_1 \psi_3\rangle \langle \psi_1 \psi_3| + (2 + |\lambda_{2312}|^2 \\ & + \lambda_{1213} \lambda_{1323} |\psi_1 \psi_2\rangle \langle \psi_2 \psi_3| + \lambda_{1223} \lambda_{2313} |\psi_1 \psi_2\rangle \langle \psi_1 \psi_3| + \lambda_{1312} \lambda_{1223} |\psi_1 \psi_3\rangle \langle \psi_2 \psi_3| \\ & + \lambda_{2313} \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{1323} \lambda_{2312} |\psi_1 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{2313} \lambda_{2313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_3| + 3\hat{\lambda}_2 = 0 \end{aligned} \quad (111)$$

As $\hat{\lambda}_2$ has no diagonal elements there is no solution, as nothing can cancel $\hat{\lambda}_1 \wedge \hat{\lambda}_1$, thus $\hat{\lambda}_1$ can not be rank 3.

4. Let

$$\hat{\lambda}_1 = |\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| + |\psi_4\rangle \langle \psi_4| \quad (112)$$

then

$$\begin{aligned} \hat{\lambda}_2 = & \lambda_{1234} |\psi_1\psi_2\rangle \langle \psi_3\psi_4| + \lambda_{3412} |\psi_3\psi_4\rangle \langle \psi_1\psi_2| + \lambda_{1324} |\psi_1\psi_3\rangle \langle \psi_2\psi_4| + \lambda_{2413} |\psi_2\psi_4\rangle \langle \psi_1\psi_3| \\ & + \lambda_{1423} |\psi_1\psi_4\rangle \langle \psi_2\psi_3| + \lambda_{2314} |\psi_2\psi_3\rangle \langle \psi_1\psi_4|. \end{aligned} \quad (113)$$

Again there are no diagonal elements.

$$\begin{aligned} \hat{\lambda}_2 \wedge \hat{\lambda}_2 = & \lambda_{1234} \lambda_{3412} |\psi_1\psi_2\rangle \langle \psi_3\psi_4| \wedge |\psi_3\psi_4\rangle \langle \psi_1\psi_2| + \lambda_{1324} \lambda_{2413} |\psi_1\psi_3\rangle \langle \psi_2\psi_4| \wedge |\psi_2\psi_4\rangle \langle \psi_1\psi_3| \\ & + \lambda_{1423} \lambda_{2314} |\psi_1\psi_4\rangle \langle \psi_2\psi_3| \wedge |\psi_2\psi_3\rangle \langle \psi_1\psi_4| \\ = & (\lambda_{1234} \lambda_{3412} + \lambda_{1324} \lambda_{2413} + \lambda_{1423} \lambda_{2314}) |\psi_1\psi_2\psi_3\psi_4\rangle \langle \psi_1\psi_2\psi_3\psi_4| \\ = & (|\lambda_{1234}|^2 + |\lambda_{1324}|^2 + |\lambda_{1423}|^2) |\psi_1\psi_2\psi_3\psi_4\rangle \langle \psi_1\psi_2\psi_3\psi_4| = 0. \end{aligned} \quad (114)$$

Thus

$$\lambda_{1234} = \lambda_{1324} = \lambda_{1423} = \lambda_{3412} = \lambda_{2413} = \lambda_{2314} = 0 \quad (115)$$

and

$$\hat{\lambda}_2 = 0, \quad (116)$$

so $\hat{\lambda}_1$ can not be rank 4.

5. Let

$$\hat{\lambda}_1 = |\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| + |\psi_4\rangle \langle \psi_4| + |\psi_5\rangle \langle \psi_5| \quad (117)$$

then there are no solutions.

• i.e.

$$|\psi_1\psi_j\rangle \langle \psi_1\psi_k| = |\psi_2\psi_j\rangle \langle \psi_2\psi_k| = \cdots = |\psi_\nu\psi_j\rangle \langle \psi_\nu\psi_k| = 0 \Rightarrow j \neq 1, \dots, \nu; k \neq 1, \dots, \nu \quad (118)$$

$$\hat{\lambda}_1 \wedge \hat{\lambda}_2 = \sum_{1 \leq j \leq \nu} |\psi_j\rangle \langle \psi_j| \wedge \sum_{1 \leq l \leq \binom{r}{2}} \beta_l |g_l\rangle \langle g_l| = \sum_{\substack{1 \leq l \leq \binom{r}{2} \\ 1 \leq j \leq \nu}} \beta_l |\psi_j g_l\rangle \langle \psi_j g_l| = 0 \quad (119)$$

Thus $\hat{\lambda}_1$ must be of the form

$$\hat{\lambda}_1 = |\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| + |\psi_4\rangle \langle \psi_4| \quad (120)$$

and

$$\hat{\lambda}_2 = \sum_{\substack{1 \leq k < l \leq 4 \\ 1 \leq m < n \leq 4}} \lambda_{klmn} |\psi_k \psi_l\rangle \langle \psi_m \psi_n|. \quad (121)$$

- the condition $\hat{\lambda}_1 \wedge \hat{\lambda}_1 + \hat{\lambda}_2^2 - \hat{\lambda}_2 = 0$ gives

$$\hat{\lambda}_1 \wedge \hat{\lambda}_1 = 2(|\psi_1 \psi_2\rangle \langle \psi_1 \psi_2| + |\psi_1 \psi_3\rangle \langle \psi_1 \psi_3| + |\psi_1 \psi_4\rangle \langle \psi_1 \psi_4| + |\psi_2 \psi_3\rangle \langle \psi_2 \psi_3| + |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4| + |\psi_3 \psi_4\rangle \langle \psi_3 \psi_4|) \quad (122)$$

which is a projector thus

$$\hat{\lambda}_2 - \hat{\lambda}_2^2 = \hat{\lambda}_2 (P_{\mathcal{H}^2} - \hat{\lambda}_2) \quad (123)$$

must also be a projector with eigenvalues $\{0, 1\} = \{\alpha_j (1 - \alpha_j)\}$ i.e.

$$1 = \alpha_j - \alpha_j^2 \quad (124)$$

- the condition $\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0$ gives

$$Tr \left\{ \hat{\lambda}_2 \wedge \hat{\lambda}_2 \right\} = \sum_{1 \leq P < Q \leq \binom{r}{2}} \alpha_P \alpha_Q = 0 \quad (125)$$

$$2 \sum_{1 \leq l < m \leq \binom{r}{2}} \beta_l \beta_m |g_l \wedge g_m\rangle \langle g_l \wedge g_m| + \sum_{1 \leq l \leq \binom{r}{2}} \beta_l^2 |g_l^2\rangle \langle g_l^2| = 0 \quad (126)$$

- if $\hat{\lambda}_2 \geq 0$ then

$$\text{rank} \left\{ \hat{\lambda}_2 \right\} = 1 \quad (127)$$

i.e.

$$\hat{\lambda}_2 = |g\rangle \langle g|. \quad (128)$$

Now again using $\hat{\lambda}_2 \wedge \hat{\lambda}_2 = 0$ gives

$$|g\rangle \langle g| \wedge |g\rangle \langle g| = |g \wedge g\rangle \langle g \wedge g| = 0, \quad (129)$$

thus

$$|g \wedge g\rangle = 0. \quad (130)$$

Utilizing the canonical expansion of $|g\rangle$ gives

$$\sum_{1 \leq l_1 < l_2 \leq s} c_{l_1} c_{l_2} |\psi_{l_1} \psi_{l_1+s} \psi_{l_2} \psi_{l_2+s}\rangle = 0 \quad (131)$$

which shows that

$$\text{rank} \{|g\rangle\} = 1 \quad (132)$$

IV. IDEMPOTENCY OF λ_2

Expanding the operator λ_2 the idempotency condition eq. (??) becomes

$$\sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l = \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} g_{l_1}^\dagger g_{l_1} g_{l_2}^\dagger g_{l_2} \quad (133)$$

and one must compare the left hand side of eq. (133) to the right hand side. In order to do this one brings the RHS of eq. (133) to normal form producing

$$\sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} g_{l_1}^\dagger g_{l_1} g_{l_2}^\dagger g_{l_2} = \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} \left\{ g_{l_1}^\dagger \left[g_{l_1}, g_{l_2}^\dagger \right] g_{l_2} + g_{l_1}^\dagger g_{l_2}^\dagger g_{l_1} g_{l_2} \right\}, \quad (134)$$

which can be developed by noting that

$$\left[g_{l_1}, g_{l_2}^\dagger \right] = \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \left[a_{j_2} a_{j_1}, a_{k_1}^\dagger a_{k_2}^\dagger \right] = -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \left[a_{k_1}^\dagger a_{k_2}^\dagger, a_{j_2} a_{j_1} \right], \quad (135)$$

that after using the commutation relationship

$$\left[a_{k_1}^\dagger a_{k_2}^\dagger, a_{j_2} a_{j_1} \right] = I_{\mathcal{H}_{\mathcal{F}}} \delta_{j_2 k_1} \delta_{j_1 k_2} - I_{\mathcal{H}_{\mathcal{F}}} \delta_{j_2 k_2} \delta_{j_1 k_1} + a_{k_2}^\dagger a_{j_2} \delta_{j_1 k_1} - a_{k_1}^\dagger a_{j_2} \delta_{j_1 k_2} - a_{k_2}^\dagger a_{j_1} \delta_{j_2 k_1} + a_{k_1}^\dagger a_{j_1} \delta_{j_2 k_2} \quad (136)$$

gives

$$\left[g_{l_1}, g_{l_2}^\dagger \right] = -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \left\{ I_{\mathcal{H}_{\mathcal{F}}} \delta_{j_2 k_1} \delta_{j_1 k_2} - I_{\mathcal{H}_{\mathcal{F}}} \delta_{j_2 k_2} \delta_{j_1 k_1} \right. \\ \left. a_{k_2}^\dagger a_{j_2} \delta_{j_1 k_1} - a_{k_1}^\dagger a_{j_2} \delta_{j_1 k_2} - a_{k_2}^\dagger a_{j_1} \delta_{j_2 k_1} + a_{k_1}^\dagger a_{j_1} \delta_{j_2 k_2} \right\},$$

then

$$= -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \left\{ \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \delta_{j_2 k_1} \delta_{j_1 k_2} - \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \delta_{j_2 k_2} \delta_{j_1 k_1} + \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \delta_{j_1 k_1} a_{k_2}^\dagger a_{j_2} \right. \quad (137)$$

$$\left. - \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \delta_{j_1 k_2} a_{k_1}^\dagger a_{j_2} - \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \delta_{j_2 k_1} a_{k_2}^\dagger a_{j_1} + \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 k_2} \delta_{j_2 k_2} a_{k_1}^\dagger a_{j_1} \right\}, \quad (138)$$

then

$$= \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \bar{g}_{l_1 j_1 j_2} g_{l_2 j_1 j_2} - \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_2 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 j_1 k_2} a_{k_2}^\dagger a_{j_2} + \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 j_1} a_{k_1}^\dagger a_{j_2} \quad (139)$$

$$+ \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_2 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 j_2 k_2} a_{k_2}^\dagger a_{j_1} - \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1 \leq r}} \bar{g}_{l_1 j_1 j_2} g_{l_2 k_1 j_2} a_{k_1}^\dagger a_{j_1}, \quad (140)$$

then

$$= \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \bar{g}_{l_1 j_1 j_2} g_{l_2 j_1 j_2} - \frac{1}{4} \sum_{\substack{1 \leq j, j_2 \leq r \\ 1 \leq k_2 \leq r}} \{ \bar{g}_{l_1 j j_2} g_{l_2 j k_2} - \bar{g}_{l_1 j j_2} g_{l_2 k_2 j} - \bar{g}_{l_1 j_2 j} g_{l_2 j k_2} + \bar{g}_{l_1 j_2 j} g_{l_2 k_2 j} \} a_{k_2}^\dagger a_{j_2}, \quad (141)$$

then

$$= \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \bar{g}_{l_1 j_1 j_2} g_{l_2 j_1 j_2} - \frac{1}{4} \sum_{\substack{1 \leq j, j_2 \leq r \\ 1 \leq k_2 \leq r}} \{ -\bar{g}_{l_1 j_2 j} g_{l_2 j k_2} - g_{l_2 k_2 j} \bar{g}_{l_1 j j_2} - \bar{g}_{l_1 j_2 j} g_{l_2 j k_2} - g_{l_2 k_2 j} \bar{g}_{l_1 j j_2} \} a_{k_2}^\dagger a_{j_2}, \quad (142)$$

hen

$$= \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \bar{g}_{l_1 j_1 j_2} g_{l_2 j_1 j_2} + \frac{1}{2} \sum_{\substack{1 \leq j, j_2 \leq r \\ 1 \leq k_2 \leq r}} \{ \bar{g}_{l_1 j_2 j} g_{l_2 j k_2} + g_{l_2 k_2 j} \bar{g}_{l_1 j j_2} \} a_{k_2}^\dagger a_{j_2}, \quad (143)$$

then

$$\delta_{l_1 l_2} + \sum_{1 \leq j_2, k_2 \leq r} \eta_{l_1 l_2 k_2 j_2} a_{k_2}^\dagger a_{j_2}, \quad (144)$$

where

$$\eta_{l_1 l_2 k_2 j_2} = \frac{1}{2} (\bar{g}_{l_1 j_2 j} g_{l_2 j k_2} + g_{l_2 k_2 j} \bar{g}_{l_1 j j_2}) = \bar{g}_{l_1 j_2 j} g_{l_2 j k_2}. \quad (145)$$

Let the rank of the operator λ_2 be two then

$$\eta_{11k_2j_2} = \bar{g}_{1j_2j}g_{1jk_2} \quad (146)$$

$$\eta_{12k_2j_2} = \bar{g}_{1j_2j}g_{2jk_2} \quad (147)$$

$$\eta_{21k_2j_2} = \bar{g}_{2j_2j}g_{1jk_2} \quad (148)$$

$$\eta_{22k_2j_2} = \bar{g}_{2j_2j}g_{2jk_2}. \quad (149)$$

The idempotency condition then becomes

$$\begin{aligned} \sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l &= \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} g_{l_1}^\dagger g_{l_1} g_{l_2}^\dagger g_{l_2} = \\ &= \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} \left\{ g_{l_1}^\dagger \left\{ \delta_{l_1 l_2} + \sum_{1 \leq j_2, k_2 \leq r} \eta_{l_1 l_2 k_2 j_2} a_{k_2}^\dagger a_{j_2} \right\} g_{l_2} + g_{l_1}^\dagger g_{l_2}^\dagger g_{l_1} g_{l_2} \right\}, \end{aligned} \quad (150)$$

giving

$$\sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l^2 g_l^\dagger g_l + \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} \left\{ \sum_{1 \leq j_2, k_2 \leq r} \eta_{l_1 l_2 k_2 j_2} g_{l_1}^\dagger a_{k_2}^\dagger a_{j_2} g_{l_2} + g_{l_1}^\dagger g_{l_2}^\dagger g_{l_1} g_{l_2} \right\}, \quad (151)$$

thus

$$\sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l |g_\kappa\rangle = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l^2 g_l^\dagger g_l |g_\kappa\rangle \iff \beta_\kappa (\beta_\kappa - 1) = 0; \quad 1 \leq \kappa \leq \binom{r}{2}. \quad (152)$$

This gives

$$\sum_{\sigma \in \{1, \dots, \binom{r}{2}\}} g_\sigma^\dagger g_\sigma = \sum_{\sigma \in \{1, \dots, \binom{r}{2}\}} g_\sigma^\dagger g_\sigma + \sum_{\sigma, \sigma' \in \{1, \dots, \binom{r}{2}\}} \left\{ \sum_{1 \leq j_2, k_2 \leq r} \eta_{\sigma \sigma' k_2 j_2} g_\sigma^\dagger a_{k_2}^\dagger a_{j_2} g_{\sigma'} + g_\sigma^\dagger g_{\sigma'}^\dagger g_\sigma g_{\sigma'} \right\} \quad (153)$$

so that

$$\sum_{\sigma, \sigma' \in \{1, \dots, \binom{r}{2}\}} \left\{ \sum_{1 \leq j_2, k_2 \leq r} \eta_{\sigma \sigma' k_2 j_2} g_\sigma^\dagger a_{k_2}^\dagger a_{j_2} g_{\sigma'} + g_\sigma^\dagger g_{\sigma'}^\dagger g_\sigma g_{\sigma'} \right\} = 0 \quad (154)$$

$$\sum_{\sigma, \sigma' \in \{1, \dots, \binom{r}{2}\}} \left\{ g_\sigma^\dagger \left(\sum_{1 \leq j_2, k_2 \leq r} \eta_{\sigma \sigma' k_2 j_2} a_{k_2}^\dagger a_{j_2} \right) g_{\sigma'} + g_\sigma^\dagger g_{\sigma'}^\dagger g_\sigma g_{\sigma'} \right\} = 0 \quad (155)$$

and in particular

$$\sum_{\sigma, \sigma' \in \{1, \dots, \binom{r}{2}\}} \sum_{1 \leq j_2, k_2 \leq r} \eta_{\sigma \sigma' k_2 j_2} g_\sigma^\dagger a_{k_2}^\dagger a_{j_2} g_{\sigma'} |g_\kappa \varphi\rangle = 0 \forall |g_\kappa \varphi\rangle \quad (156)$$

$$\eta_{11k_2j_2} = \sum_{1 \leq j \leq r} \bar{g}_{1j_2j} g_{1jk_2} = - \left(\mathbb{G}_1^\dagger \mathbb{G}_1 \right)_{k_2j_2} \quad (157)$$

$$\eta_{12k_2j_2} = \sum_{1 \leq j \leq r} \bar{g}_{1j_2j} g_{2jk_2} = - \left(\mathbb{G}_1^\dagger \mathbb{G}_2 \right)_{k_2j_2} \quad (158)$$

$$\eta_{21k_2j_2} = \sum_{1 \leq j \leq r} \bar{g}_{2j_2j} g_{1jk_2} = - \left(\mathbb{G}_2^\dagger \mathbb{G}_1 \right)_{k_2j_2} \quad (159)$$

$$\eta_{22k_2j_2} = \sum_{1 \leq j \leq r} \bar{g}_{2j_2j} g_{2jk_2} = - \left(\mathbb{G}_2^\dagger \mathbb{G}_2 \right)_{k_2j_2} . \quad (160)$$

$$\begin{aligned} - \sum_{1 \leq j_2, k_2 \leq r} \left\{ \left(\mathbb{G}_1^\dagger \mathbb{G}_1 \right)_{k_2j_2} g_1^\dagger a_{k_2}^\dagger a_{j_2} g_1 + \left(\mathbb{G}_1^\dagger \mathbb{G}_2 \right)_{k_2j_2} g_1^\dagger a_{k_2}^\dagger a_{j_2} g_2 \right. \\ \left. + \left(\mathbb{G}_2^\dagger \mathbb{G}_1 \right)_{k_2j_2} g_2^\dagger a_{k_2}^\dagger a_{j_2} g_1 + \left(\mathbb{G}_2^\dagger \mathbb{G}_2 \right)_{k_2j_2} g_2^\dagger a_{k_2}^\dagger a_{j_2} g_2 \right\} \quad (161) \end{aligned}$$

$$\begin{aligned} \sum_{1 \leq j, j_2, k_2 \leq r} \left\{ \bar{g}_{1j_2j} g_{1jk_2} g_1^\dagger a_{k_2}^\dagger a_{j_2} g_1 + \bar{g}_{1j_2j} g_{2jk_2} g_1^\dagger a_{k_2}^\dagger a_{j_2} g_2 + \bar{g}_{2j_2j} g_{1jk_2} g_2^\dagger a_{k_2}^\dagger a_{j_2} g_1 + \bar{g}_{2j_2j} g_{2jk_2} g_2^\dagger a_{k_2}^\dagger a_{j_2} g_2 \right\} |g_\kappa \varphi_\nu\rangle \\ = 0 \quad \forall |g_\kappa \varphi_\nu\rangle \quad (162) \end{aligned}$$

$$\sum_{1 \leq j, k_2 \leq r} \{ \bar{g}_{1\nu j} g_{1jk_2} |g_1 \varphi_{k_2}\rangle + \bar{g}_{2\nu j} g_{1jk_2} |g_2 \varphi_{k_2}\rangle \} = 0 \quad (163)$$

$$\sum_{1 \leq k_2 \leq r} \left\{ \left(\mathbb{G}_1^\dagger \mathbb{G}_1 \right)_{k_2\nu} |g_1 \varphi_{k_2}\rangle + \left(\mathbb{G}_2^\dagger \mathbb{G}_1 \right)_{k_2\nu} |g_2 \varphi_{k_2}\rangle \right\} = 0 \quad (164)$$

$$\begin{aligned} \lambda_2 &= \sum_{\substack{1 \leq j_1 < j_2 \leq \infty \\ 1 \leq k_1 < k_2 \leq \infty}} \lambda_{2j_1j_2k_1k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = \sum_{1 \leq l \leq r} \beta_l g_l^\dagger g_l \\ &= \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1, j_2 \leq s}} c_{lj_1} c_{lj_2} \beta_l b_{lj_1}^\dagger b_{lj_1+s}^\dagger b_{lj_2+s} b_{lj_2} = \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1, j_2 \leq s}} c_{lj_1} c_{lj_2} \beta_l b_{lj_1}^\dagger \left(\delta_{j_1j_2} - b_{lj_2+s} b_{lj_1+s}^\dagger \right) b_{lj_2} \\ &= \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1 \leq s}} c_{lj_1} c_{lj_1} \beta_l b_{lj_1}^\dagger b_{lj_1} - \sum_{\substack{1 \leq l \leq r \\ 1 \leq j_1, j_2 \leq s}} c_{lj_1} c_{lj_2} \beta_l b_{lj_1}^\dagger b_{lj_2+s} b_{lj_1+s}^\dagger b_{lj_2} \quad (165) \end{aligned}$$